

Question Set 1

1. Which of the following best captures the main differences between “scaling up” and “scaling out” in distributed systems?
 - a. Scaling up add more powerful hardware to a single machine.
 - b. Scaling out replaces existing nodes with faster storage devices.
 - c. Scaling up requires data replication across machines.
 - d. Scaling out increases CPU frequency of each core.

Answer: a. scaling up (vertical) means adding more power (CPU, memory) to a single machine, while scaling out adds more nodes to a cluster. L3

2. In the MapReduce model, what is the primary purpose of the shuffle and sort phase?
 - a. To partition the input data among the mappers evenly.
 - b. To aggregate intermediate key-value pairs by key before sending to reducers.
 - c. To combine map outputs into a single output file.
 - d. To execute user-defined functions on each key.

Answer: b. L3

3. Which of the following characteristics distinguishes HDFS from traditional file systems?
 - a. It stores small files and memory to reduce latency.
 - b. It provides a single-writer, multiple-reader access model.
 - c. It uses synchronous replication across all racks.
 - d. It requires manual metadata management by users.

Answer: b. Only one writer can modify a file at a time, ensuring consistency. L2

4. In BigTable, how are tablets organized and located efficiently?
 - a. Each tablet server stores its tablets in local memory only.
 - b. Tablet metadata is stored in a special METADATA table indexed through Chubby.
 - c. The master polls every tablet server to discover tablets.
 - d. Tablet locations are encoded directly in the row keys.

Answer: b. L3

5. What is the main trade-off expressed by the CAP theorem?
 - a. A distributed system can provide only two of consistency, availability, and partition tolerance.
 - b. It limits the total capacity of nodes in a cluster.
 - c. It defines how replication factor influences throughput.
 - d. It ensures atomicity of transactions in distributed databases.

Answer: a. L3

6. In Cassandra, which mechanism ensures that all replicas eventually converge to the same state?
 - a. Gossip protocol.
 - b. Read repair and anti-entropy Merkle tree comparison.
 - c. Dynamic snitching for query optimization.
 - d. Consistent hashing ring rebalance.

Answer: b. L3

7. What is the main limitation of MapReduce that motivated the design of Apache Spark?
 - a. Lack of fault tolerance for failed map tasks.
 - b. Disk-based intermediate storage causing high I/O overhead.
 - c. Inability to process structured data.
 - d. Absence of support for user-defined functions.

Answer: b. L5

8. Which statement correctly describes a Spark RDD (Resilient Distributed Dataset)?
 - a. It's a mutable distributed data structure used for random updates.
 - b. It tracks its lineage to recompute lost partitions when failures occur.
 - c. It supports SQL queries directly without transformations.
 - d. It provides guaranteed schema enforcement at runtime.

Answer: b. Lecture 5

9. In Spark, a wide transformation such as `groupByKey()` differs from a narrow transformation because it:
- Requires data shuffling across partitions.
 - Operates only on cached datasets.
 - Can be executed entirely within a single partition.
 - Is performed only in the driver process.

Answer: a. Lecture 5

10. When using Spark SQL, what is the relationship between a `DataFrame` and a `Dataset`?
- A `DataFrame` is a `Dataset` of generic Row objects.
 - A `Dataset` is a `DataFrame` without schema information.
 - `DataFrames` are typed, while `Datasets` are untyped.
 - `Datasets` cannot be queried using SQL.

Answer: a. Lecture 6

11. Which of the following best describes the function of Spark's Catalyst Optimizer?
- It manages cluster resources for executors.
 - It rewrites query plans to produce an efficient physical execution plan.
 - It performs checkpointing of intermediate RDDs.
 - It generates the user interface for the Spark UI.

Answer: b. Lecture 6

12. In Structured Streaming, what does a watermark represent?
- The maximum number of records per micro-batch.
 - A threshold defining how long to wait for late arriving events.
 - The lower bound of window aggregation intervals.
 - A signal used to trigger recomputation of missing batches.

Answer: b. Lecture 8

13. Which of the following best characterizes a tumbling window in stream processing?
- Non-overlapping, fixed-size time windows that cover the entire timeline.
 - Overlapping sliding windows used for incremental results.
 - Windows triggered by a specific event rather than time.
 - Dynamically resized windows based on input rate.

Answer: a. Lecture 8

14. In Apache Kafka, how does the system ensure message ordering?
- Messages are ordered within each partition according to their offsets.
 - A global sequence number is applied across all topics.
 - Brokers use timestamps to reorder messages at consumption time.
 - Producers enforce total ordering before sending messages.

Answer: a. Lecture 8

15. In GraphX or similar graph frameworks, what is the purpose of triplets?
- To store only vertex metadata separately from edges.
 - To represent an edge along with its source and destination vertex attributes.
 - To aggregate all edges sharing the same vertex.
 - To partition the graph based on vertex degree.

Answer: b. Lecture 10

16. Compared to Pregel, why can GraphLab's asynchronous execution achieve faster convergence?
- Vertices update immediately without waiting for global synchronization barriers.
 - GraphLab executes all vertices sequentially.
 - It uses random sampling instead of message passing.
 - It eliminates the need for edge storage.

Answer: a. Lecture 10

17. What is the primary objective of a two-level scheduler such as in Mesos or YARN?

- a. To delegate fine-grained task placement to application frameworks.
- b. To centralize all scheduling decisions in a global master.
- c. To prevent resource fragmentation by merging all jobs.
- d. To statically allocate resources before job submission.

Answer: a. The global manager allocates resources, and frameworks schedule their own tasks. Lecture 11

18. Which fairness policy ensures that users receive resources proportionally to their weights until one reaches full capacity?
- a. Proportional share fairness.
 - b. Weighted max-min fairness.
 - c. Dominant resource fairness.
 - d. Capacity-based fairness.

Answer: b. Lecture 11

19. What feature of Delta Lake enables time travel and rollback of table versions?
- a. Immutable Parquet files stored with timestamps.
 - b. A transaction log (`_delta_log`) recording every data modification.
 - c. A snapshot cache stored in memory.
 - d. Periodic replication of data directories.

Answer: b. Lecture 12

20. Which of the following statements about a data lakehouse is true?
- a. It combines the low-cost storage of data lakes with ACID transactions and schema enforcement from data warehouses.
 - b. It is an in-memory database used exclusively for analytics.
 - c. It replaces ETL processes with direct sensor ingestion.
 - d. It stores only structured data in columnar format.

Answer: a. Lecture 12